Qatar-1b

R-0240

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_180523` · created 2026-07-17T05:04:29+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458261.896 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.62 +/- 2.14 [%]
Orbital Inclination (inc)	86.44 +/- 2.93
Airmass coefficient 1 (a1)	1.25 +/- 0.46
Airmass coefficient 2 (a2)	-0.17 +/- 0.57
Scatter in the residuals of the lightcurve fit is	35.76 %
Best Comparison Star	#1 - [141, 283]
Optimal Aperture	1.82
Optimal Annulus	3.0
Transit Duration (day)	0.077 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.056 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.077 d vs 0.0692 d (deviation 0.6σ)
Mid-time O–C	-2.9 min
Depth SNR	3.4
Residual scatter	35.76 %

Light curve

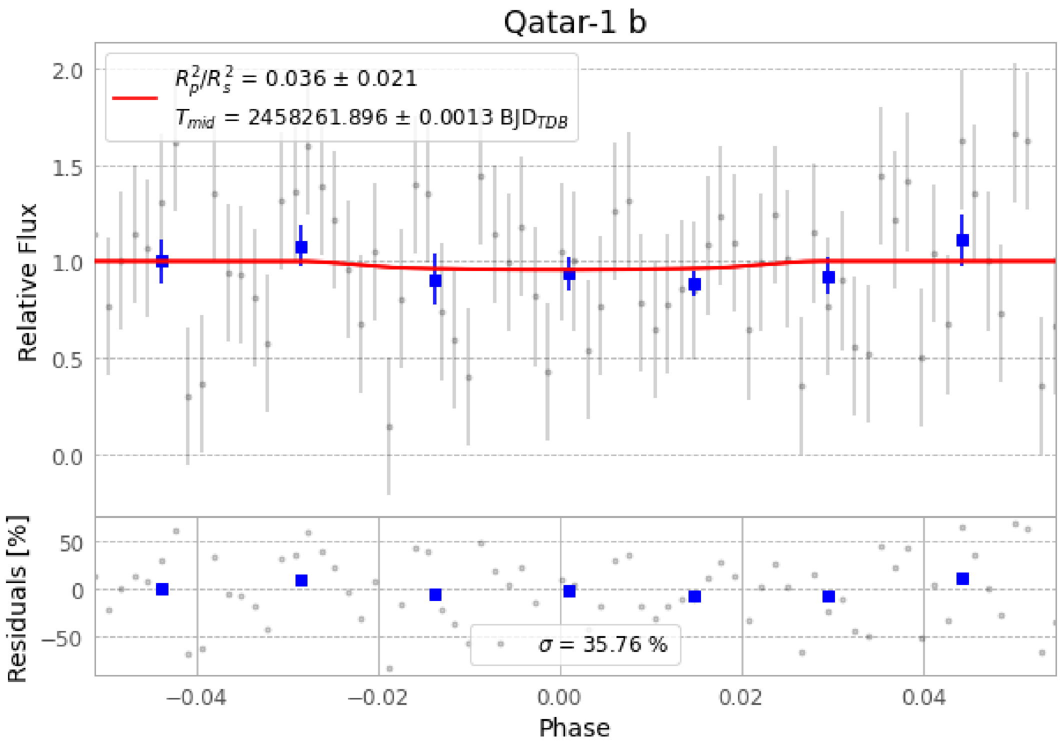

Figure 1 — FinalLightCurve_Qatar-1 b_2018-05-23.png (SHA-256 89a0585a64a8eff6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [121, 345], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8921438e80637be2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e`

Data provenance

73 FITS frames (48 MB), `Qatar-1180523074512.FITS` → `Qatar-1180523112112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180523.log (5eba7b1d53e0…), FinalLightCurve_Qatar-1 b_2018-05-23.png (89a0585a64a8…), FinalLightCurve_Qatar-1 b_2018-05-23.csv (4881240ca18f…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)